

Predicting Readmission Rate for Diabetic Patients Using Machine Learning

Team Members and Emails

Colin Frishberg (cpfrish@berkeley.edu), Terra Jiang (yjiang66@berkeley.edu),
Sai Sriya Mudigonda (smudigonda@berkeley.edu), Rahil Sharma (rahilsharma@berkeley.edu)

Motivation

The hospital readmission rate is a major concern for both healthcare providers and patients, as it reflects the balance between reducing hospital costs and maintaining treatment efficiency and effectiveness. Among chronic conditions, diabetes is particularly challenging with complicated disease treatments and high readmission rates due to insufficient follow-up care and complex discharge procedures. Therefore, predicting which patients are likely to be readmitted soon after discharge has become a significant clinical focus and research priority. To address this, hospitals often target high-risk diabetes patients with interventions such as patient education programs, medication adjustments, or closer follow-up appointments to improve quality of care and reduce preventable readmissions.

The Centers for Medicare and Medicaid Services (CMS) launched the Hospital Readmissions Reduction Program (HRRP) on October 1, 2012. According to this program, hospitals that fail to control readmission rates face not only financial penalties but also public scrutiny. In this project, we aim to utilize machine learning to develop a predictive model that can identify patients at risk of complications before discharge. In addition to cost reduction and compliance with CMS regulations, our goal is to uncover which patient demographics (e.g., weight, height, age, ...), dosage changes, and other hospital-related features contribute the most to the likelihood of readmissions.

Data

The dataset used in this project originates from the University of California, Irvine (UCI) Machine Learning Repository (*Diabetes*), a well-known source for benchmark datasets in data science research. This dataset is widely studied in healthcare analytics as it provides valuable insights into risk factors and predictors of hospital readmission among diabetic patients. It contains 101,766 patient records collected between 1998 and 2008 across 130 different hospitals and integrated delivery networks in the United States. Each record corresponds to a patient diagnosed with diabetes who received laboratory tests, was prescribed medication, and experienced a hospital stay of up to 14 days. In total, the dataset provides 47 input features spanning both categorical and numerical variables. Our main focus will be on the binary target variable, *readmission*, which indicates whether or not a patient was readmitted to the hospital.

Related work

Prior studies in this research domain have demonstrated that machine learning can effectively predict hospital readmissions, with key predictors including prior inpatient visits, discharge disposition, admission type, and medication use. While these works highlight opportunities for cost savings and improved resource allocation, they primarily focus on identifying readmission risk without delving into the impact of interactions between demographic and hospital-related variables on outcomes. Building on this foundation, our project will not only develop a predictive model for complications prior to discharge but also identify whether patient demographics, dosage changes, and clinical features contribute to readmission likelihood, supporting both cost reduction and compliance with CMS regulations.

Project GitHub Repo

https://github.com/UC-Berkeley-I-School/w207_final_project_frishberg_jiang_mudigonda_sharma

References

- Hospital Readmissions Reduction Program* | CMS. (n.d.).
[https://www.cms.gov/medicare/payment/prospective-payment-systems/acute-inpatient-pps/hospital-readmissions-reduction-program-hrrp](https://www.cms.gov/medicare/payment/prospective-payment-systems/acute-inpatientpps/hospital-readmissions-reduction-program-hrrp)
- Diabetes 130-US hospitals for years 1999-2008 - UCI Machine Learning Repository*. (n.d.).
<https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008>
- Sharda, M., Sharma, S., Raikar, S., Verhagen, N., Wagle, J., Mathur, R., Gowda, S., Kommu, S., Prasad, R., Bhandari, S., & Jha, P. (2025). The Role of machine Learning in Predicting hospital readmissions among general internal medicine Patients: A Systematic review. *Cureus*. <https://doi.org/10.7759/cureus.84761>
- Bhuvan, M. S., Kumar, A., Zafar, A., & Kishore, V. (2016, February 12). *Identifying Diabetic Patients with High Risk of Readmission*. arXiv.org. <https://arxiv.org/abs/1602.04257>
- Liu, V. B., Sue, L. Y., & Wu, Y. (2024). Comparison of machine learning models for predicting 30-day readmission rates for patients with diabetes. *Journal of Medical Artificial Intelligence*, 7, 23. <https://doi.org/10.21037/jmai-24-70>